

will then increase it approaches point Q.

14. Answer: C

$$g = \frac{GM_{\text{Saturn}}}{r^2} = r\omega^2$$

$$\frac{GM_{\text{Saturn}}}{r^3} = \left(\frac{2\pi}{T}\right)^2$$

$$M_{\text{Saturn}} = \frac{4\pi^2 (2.95 \times 10^8)^3}{G(1.89 \times 24 \times 3600)^2} = 5.70 \times 10^{26} \text{ kg}$$